Seoul, Korea
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Minwoo Jung

EDUCATION

Seoul National University, Republic of Korea *Sep. 2023 - Present*

- **Ph.D. Candidate** in Mechanical Engineering
- Cumulative GPA 3.71/4.3
- Robust Perception and Mobile Robotics Lab, advised by Prof. Ayoung Kim

Seoul National University, Republic of Korea *Sep. 2021 - Aug. 2023*

- **M.S.** in Mechanical Engineering
- Cumulative GPA 4.06/4.3
- Thesis: "Asynchronous Multiple LiDAR-Inertial Odometry using Point-wise Inter-LiDAR Uncertainty Propagation"
- Robust Perception and Mobile Robotics Lab, advised by Prof. Ayoung Kim

KAIST, Republic of Korea *Mar. 2021 - Aug. 2021*

- **M.S. Student** in Civil and Environmental Engineering
- Moved to Seoul National University
- Advised by Prof. Ayoung Kim

KAIST, Republic of Korea *Mar. 2017 - Feb. 2021*

- **B.S.** in Civil and Environmental Engineering
- Cumulative GPA 3.76/4.3
- Graduated *cum laude*

RESEARCH INTERESTS

- Range Sensor, SLAM, Global Localization, Sensor Fusion, World Model, and Field Robotics

LANGUAGES & SKILLS

- Korean (native), English
- Programming: C/C++, Python, MATLAB, PyTorch
- Robotics: Multi-LiDAR SLAM, Sensor Fusion, Global Localization, Autonomous Navigation
- Tools: ROS, GTSAM, SymForce, Docker, Linux, LaTeX

RESEARCH EXPERIENCE

Collaborative Research with Oxford Robotics Institute DRS Group *2024 - Present*

- Visiting Research Student at Oxford Robotics Institute, Feb. 2025 - Mar. 2025.
- Implemented LiDAR place recognition using vision foundation models.
- Implemented forest localization using inter-tree geometric matching.

Global Re-localization for LiDAR SLAM *2023 - 2024*

- Developed global re-localization for scan-to-submap registration in GPS-denied unstructured environments.
- Integrated the module into an existing SLAM framework and validated it with real-world multi-session data.

Multiple LiDAR-IMU SLAM *2022 - 2023*

- Developed a multiple LiDAR-IMU SLAM framework for trajectory estimation in unstructured environments.
- Integrated the module into a hardware platform and validated it in real-world wild environments.

Autonomous Navigation in Construction Sites *2021 - 2025*

- Implemented an autonomous navigation module for a mobile ground robot equipped with LiDAR, IMU, and GPS.
- Demonstrated the system across multiple construction sites.

PUBLICATIONS

International Journal

1. Minwoo Jung, Dongjae Lee, Nived Chebrolu, Haedam Oh, Maurice Fallon, and Ayoung Kim. TreeLoc++: Robust 6-DoF LiDAR Localization in Forests with a Compact Digital Forest Inventory. *IEEE Transactions on Field Robotics* (*T-FR*), 2026.
2. Wooseong Yang, Dongjae Lee, Minwoo Jung, and Ayoung Kim. Geometrically-Constrained Radar-Inertial Odometry via Continuous Point-Pose Uncertainty Modeling. *IEEE Robotics and Automation Letters* (*RA-L*), 11(5):6098–6105, 2026.
3. Hyesu Jang, Wooseong Yang, Hanguen Kim, Dongje Lee, Yongjin Kim, Jinbum Park, Minsoo Jeon, Jaeseong Koh, Yejin Kang, Minwoo Jung, Sangwoo Jung, and Ayoung Kim. MOANA: Multi-Radar Dataset for Maritime Odometry and Autonomous Navigation Application. *The International Journal of Robotics Research* (*IJRR*), 45(2):193–203, 2026.
4. Hanjun Kim, Minwoo Jung, Wooseong Yang, and Ayoung Kim. SHeRLoc: Synchronized Heterogeneous Radar Place Recognition for Cross-Modal Localization. *IEEE Robotics and Automation Letters* (*RA-L*), 10(12):13264–13271, 2025.
5. Minwoo Jung, Wooseong Yang, Dongjae Lee, Hyeonjae Gil, Giseop Kim, and Ayoung Kim. HeLiPR: Heterogeneous LiDAR Dataset for inter-LiDAR Place Recognition under Spatiotemporal Variations. *The International Journal of Robotics Research* (*IJRR*), 43(12):1867–1883, 2024.
6. Dongjae Lee, Minwoo Jung, Wooseong Yang, and Ayoung Kim. LiDAR Odometry Survey: Recent Advancements and Remaining Challenges. *Intelligent Service Robotics* (*ISR*), 17:95–118, 2024.
7. Hyesu Jang, Minwoo Jung, Myung-Hwan Jeon, and Ayoung Kim. LodeStar: Maritime Radar Descriptor for Semi-Direct Radar Odometry. *IEEE Robotics and Automation Letters* (*RA-L*), 9(2):1684–1691, 2024.
8. Sangwoo Jung, Hyesu Jang, Minwoo Jung, Ayoung Kim, and Myung-Hwan Jeon. Imaging Radar and LiDAR Image Translation for 3-DOF Extrinsic Calibration. *Intelligent Service Robotics* (*ISR*), 17:167–179, 2024.
9. Jeongyun Kim, Myung-Hwan Jeon, Sangwoo Jung, Wooseong Yang, Minwoo Jung, Jaeho Shin, and Ayoung Kim. TRansPose: Large-Scale Multispectral Dataset for Transparent Object. *The International Journal of Robotics Research* (*IJRR*), 43(6):731–738, 2024.
10. Minwoo Jung, Sangwoo Jung, and Ayoung Kim. Asynchronous Multiple LiDAR-Inertial Odometry using Point-wise Inter-LiDAR Uncertainty Propagation. *IEEE Robotics and Automation Letters* (*RA-L*), 8(7):4211–4218, 2023.
11. Jeongyun Kim, Seungsang Yun, Minwoo Jung, Ayoung Kim, and Younggun Cho. Automatic Wall Slant Angle Map Generation using 3D Point Clouds. *ETRI Journal*, 43(4):594–602, 2021.

Domestic Journal

1. Eunho Lee, Minwoo Jung, Jongho Kim, Kyongsu Yi, and Ayoung Kim. Robust 3D Object Detection through Distance based Adaptive Thresholding. *The Journal of Korea Robotics Society*, 19(1):106–116, 2024.
2. Sangwoo Jung, Minwoo Jung, and Ayoung Kim. Map Error Measuring Mechanism Design and Algorithm Robust to LiDAR Sparsity. *The Journal of Korea Robotics Society*, 16(3):189–198, 2021.
3. Minwoo Jung, Sangwoo Jung, Hyesu Jang, and Ayoung Kim. Intensity and Ambient Enhanced LiDAR-Inertial SLAM for Unstructured Construction Environment. *The Journal of Korea Robotics Society*, 16(3):179–188, 2021.

International Conference Proceedings

1. Geonhui Lee, Hyeonjae Gil, Ayoung Kim, and Minwoo Jung. ELite++: Efficient Lifelong LiDAR Mapping via Window-Based Registration and Voxel Ratio Maps. *International Conference on Control, Automation and Systems* (*ICCAS*), 2026.
2. Minwoo Jung, Nived Chebrolu, Lucas Carvalho de Lima, Haedam Oh, Maurice Fallon, and Ayoung Kim. TreeLoc: 6-DoF LiDAR Global Localization in Forests via Inter-Tree Geometric Matching. *IEEE International Conference on Robotics and Automation* (*ICRA*), 2026.
3. Minwoo Jung, Lanke Frank Tarimo Fu, Maurice Fallon, and Ayoung Kim. ImLPR: Image-based LiDAR Place Recognition using Vision Foundation Models. *Conference on Robot Learning* (*CoRL*), pages 3318–3340, 2025.
4. Chiyun Noh, Wooseong Yang, Minwoo Jung, Sangwoo Jung, and Ayoung Kim. GaRLIO: Gravity enhanced Radar-LiDAR-Inertial Odometry. *IEEE International Conference on Robotics and Automation* (*ICRA*), pages 9869–9875, 2025.
5. Hanjun Kim, Minwoo Jung, Chiyun Noh, Sangwoo Jung, Hyunho Song, Wooseong Yang, Hyesu Jang, and Ayoung Kim. HeRCULES: Heterogeneous Radar Dataset in Complex Urban Environment for Multi-session Radar SLAM. *IEEE International Conference on Robotics and Automation* (*ICRA*), pages 4649–4656, 2025.
6. Minwoo Jung, Sangwoo Jung, Hyeonjae Gil, and Ayoung Kim. HeLiOS: Heterogeneous LiDAR Place Recognition via Overlap-based Learning and Local Spherical Transformer. *IEEE International Conference on Robotics and Automation* (*ICRA*), pages 2204–2211, 2025.
7. Sanghyun Hahn, Seunghun Oh, Minwoo Jung, Ayoung Kim, and Sangwoo Jung. Quantitative 3D Map Accuracy Evaluation Hardware and Algorithm for LiDAR (-Inertial) SLAM. *International Conference on Control, Automation and Systems* (*ICCAS*), pages 568–572, 2024.
8. Hyesu Jang, Minwoo Jung, and Ayoung Kim. RaPlace: Place Recognition for Imaging Radar using Radon Transform and Mutable Threshold. *IEEE/RSJ International Conference on Intelligent Robots and Systems* (*IROS*), pages 11194–11201, 2023.
9. Seungsang Yun, Minwoo Jung, Jeongyun Kim, Sangwoo Jung, Younghun Cho, Myung-Hwan Jeon, Giseop Kim, and Ayoung Kim. STheReO: Stereo Thermal Dataset for Research in Odometry and Mapping. *IEEE/RSJ International Conference on Intelligent Robots and Systems* (*IROS*), pages 3857–3864, 2022.

Preprints and Workshops

1. Beomsu Kim, Minwoo Jung, and Giseop Kim. Are Visual Place Recognition Models Recognizing Places or Conditions? Distractor-Augmented Evaluation and Condition Suppression. *arXiv preprint arXiv:2608.06847*, 2026.
2. Hyunho Song, Dongjae Lee, Seunghun Oh, Minwoo Jung, and Ayoung Kim. The City that Never Settles: Simulation-based LiDAR Dataset for Long-Term Place Recognition Under Extreme Structural Changes. *Construction Robotics Workshop, ICRA*, 2025.
3. Eunho Lee, Minwoo Jung, and Ayoung Kim. Toward Robust LiDAR based 3D Object Detection via Density-Aware Adaptive Thresholding. *Field Robotics Workshop, ICRA*, 2024.
4. Dongjae Lee, Minwoo Jung, and Ayoung Kim. ConPR: Ongoing Construction Site Dataset for Place Recognition. *Closing the Loop on Localization Workshop, IROS*, 2024.
5. Minwoo Jung, Dongjae Lee, Wooseong Yang, and Ayoung Kim. HeLiPR: Heterogeneous LiDAR Dataset for inter-LiDAR Place Recognition in Spatial and Temporal Variation Environments. *Novel Sensors Workshop, IROS*, 2023.
6. Minwoo Jung, Sangwoo Jung, and Ayoung Kim. MA-LIO: Multiple Asynchronous LiDAR-Inertial Odometry with Time-dependent Point-wise Uncertainty. *Towards Safe Autonomy Workshop, Robotics: Science and Systems* (*RSS*), 2023.
7. Hyesu Jang, Minwoo Jung, Sangwoo Jung, and Ayoung Kim. Radar-based Angular Correction for Tightly-coupled LiDAR Odometry. *IROS Workshop on Perception and Navigation for Autonomous Robotics in Unstructured and Dynamic Environments*, 2022.
8. Minwoo Jung and Ayoung Kim. Pointclouds Integration from Aerial and Ground View Exploiting Normal Vector and Pose Graph Optimization. *ICRA Late Breaking Results*, 2022.

TEACHING & TUTORIALS

Tutorial: ICP and LiDAR Localization *Apr. 2026*
Multidisciplinary Creative Class

Tutorial: Understanding 3D LiDAR Point Clouds and Their Practical Usage *Mar. 2026*
Machine Intelligence Industry-Academia Open Lecture and Consulting

Tutorial: LiDAR Point Cloud Matching *Nov. 2025*
[M3228.000300] Sensor-Based Spatial Intelligence, Seoul National University

Head Teaching Assistant, [M3228.000300] Sensor-Based Spatial Intelligence *Sep. 2025*
Seoul National University

PROFESSIONAL SERVICE

- Reviewer: The International Journal of Robotics Research (IJRR)
- Reviewer: The International Conference on Control, Automation and Systems (ICCAS)
- Reviewer: The International Journal of Digital Earth (IJDE)
- Reviewer: IEEE Transactions on Robotics (T-RO)
- Reviewer: IEEE Robotics and Automation Letters (RA-L)
- Reviewer: IEEE International Conference on Robotics and Automation (ICRA)
- Reviewer: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- Reviewer: Conference on Neural Information Processing Systems (NeurIPS)
- Reviewer: Conference on Robot Learning (CoRL)
- Reviewer: IEEE International Conference on Ubiquitous Robots (UR)
- Reviewer: IEEE Transactions on Automation Science and Engineering (T-ASE)

AWARDS AND HONORS

2025 Paper Award (2nd author), Korea Robotics Society (KROS)
2023 Best Overall Presentation Award (2nd author), IROS Workshop
2023 Best M.S. Thesis Award, Seoul National University
2023 Ranked 4th in the LiDAR Single Session category, Hilti SLAM Challenge
2021 Graduation Citation, Korea Concrete Institute
2021 Development Foundation Chairman Award, KAIST
2021 Cum Laude, Civil and Environmental Engineering, KAIST
2019 Dean's List, Civil and Environmental Engineering, KAIST